DEEPP Project

Customer Credit Profile Prediction Model

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Executive Summary

Project Description:

- Older credit prediction model are losing their accuracy and the bank want a new prediction model
- What are the key feature/variable when deciding on a good/bad credit card client?
- What kind of demographic and income type that tends to be a good credit card client?

Objective:

- Identify patterns and key features indicating default risk among credit card applicants, achieving a predictive model with at least 85% accuracy within four weeks, while analyzing demographic and income characteristics of successful clients.

Tools:

- Python
- Tableau

Metrics:

- Metrics used are Income Amount, Work Year, Age, Income Type, Flagging Own a Realty or Car, and Education Level

Analysis Result:

- 3 Key Feature that effect the prediction model are Income amount, Age, and Work year.
- Supporting Feature that have an effect in the prediction model are Income Type, Flagging Own a Realty or Car, and Education Level.

Recommendations:

- Focus on the Key Feature
- Check on Supporting Feature
- Cater to Clients Needs

Background Project

- Credit score cards are utilized in financial institutions for risk control, predicting future defaults and borrowing behaviors based on applicant data and historical records.

- Older credit scoring model uses machine learning model such as logistics model which gives banks transparency in terms of feature impact because they the result can be scaled up and easier to interpret.

- Faced with large economic fluctuations, older model have diminishing accuracy of its prediction. Which is why the bank needed a new machine learning model to make accurate predictions

Problem Statement and Objective

Problem Statement

- What are the key feature/variable when deciding on a good/bad credit card client?
- What kind of demographic and income type that tends to be a good credit card client?

Hypothesis

- The larger the yearly income of a client the lower the risk of a credit default by the client.
- Client with a stable income / working have a lower risk in credit default than client that is not working anymore.
- Client that is older are more likely to have lower risk of credit Default

Project Objective

- Identify patterns and key features indicating default risk among credit card applicants, achieving a predictive model with at least 85% accuracy within four weeks, while analyzing demographic and income characteristics of successful clients.

Methodology

Defining the Problem:

- Data understanding and explore initial data
- Defining key problems and also stakeholder of the project

Data Cleaning

- Data Cleaning are fully done using python
- Search and fix missing data, and duplicates

Data Analysis

- Data analysis used in this project is EDA and Random Forest Machine Learning

Insight Generations

- Using the result of the analysis generate insight

Recommendations

- From the insight generate recommendations

Data Sets and Scope

Account Records

This Data sets contains all of the account of the banks currently both containing credit card clients and non credit card clients

DATA SET LINK

Credit Records

This Data sets Contain all of the past credit records of all the credit card users including the status and which month

DATA SET LINK

Data Scope

The scope of the data used in for the projects are all of the data from both the account records and also the credit records data sets but we will only use data that have a match of ID in both data sets.

Machine Learning Model with the Most Accuracy are XG Boost and Random Forest

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.530663	0.530669	0.530663	0.530641
K-Neighbors Classifier	0.703066	0.723857	0.703066	0.696008
Decision Tree Classifier	0.816959	0.831347	0.816959	0.814950
Random Forest Classifier	**0.816959**	0.833806	0.816959	0.814620
XGBoost Classifier	**0.881727**	0.886970	0.881727	0.881325

- As previously mentioned in the problem statement, prediction model using logistics regression have the lowest accuracy in predicting a good or bad credit score as time goes by.

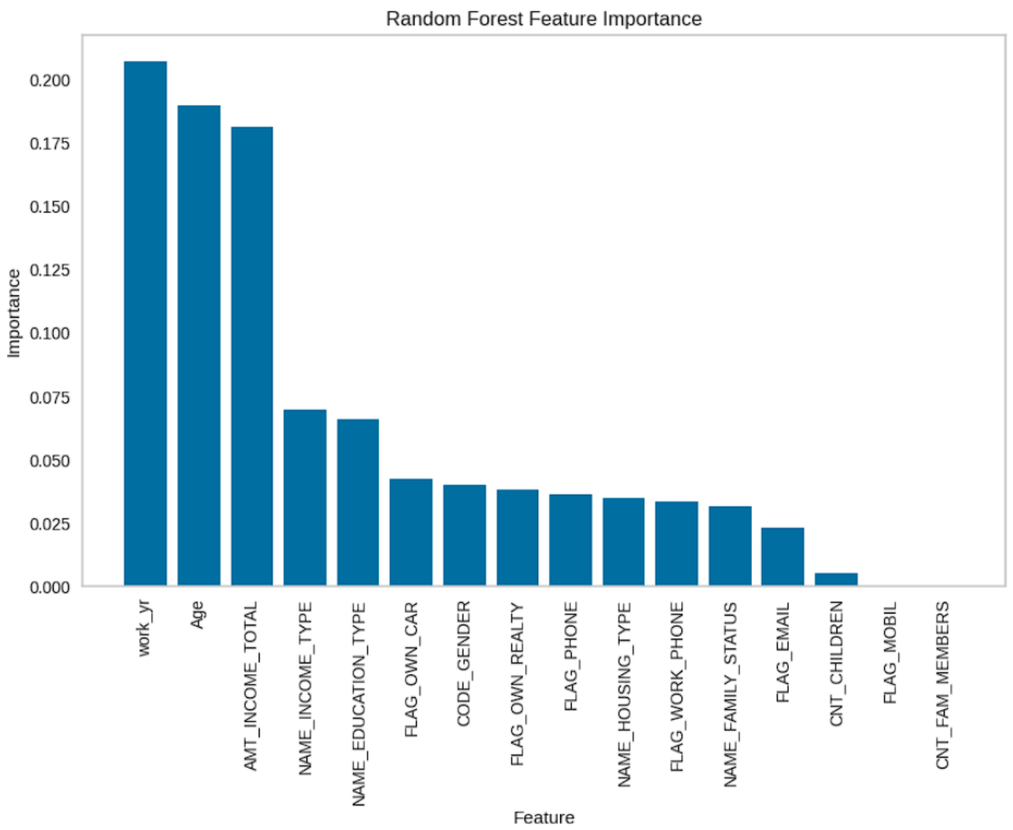
- Although XG Boost Classifier model has a higher accuracy than the random forest model, XG boost by it self function to boost its classifier that is more complicated and less transparent when it comes to interpret it. That is why we will use Random Forest model as the Machine learning model for the credit score prediction model.

Work Year, Age, and Yearly Income Amount are the 3 Main Key Features of the Prediction Model

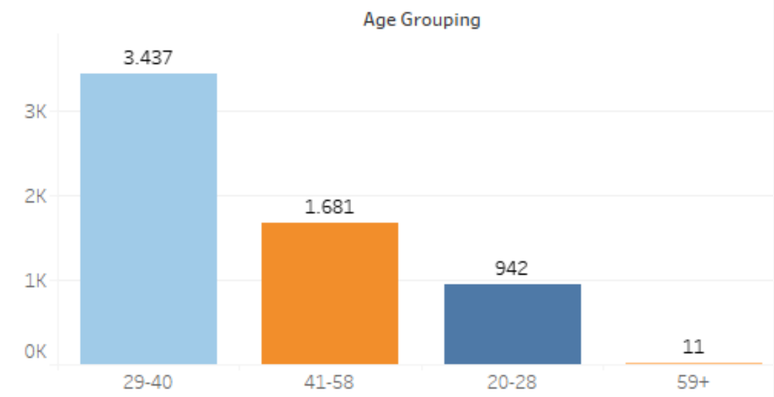
After the machine learning model has been chosen then it is time to see which of all the feature used in the data set is the key feature / variable in deciding a clients credit worthiness which can be seen in the feature importance.

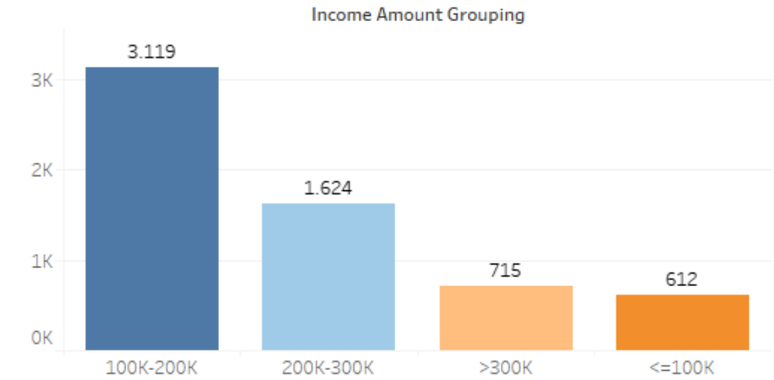
- Based on the prediction model that has been created there is 3 main feature in deciding a good or bad credit client, which are **work year, age, and amount of yearly income**.

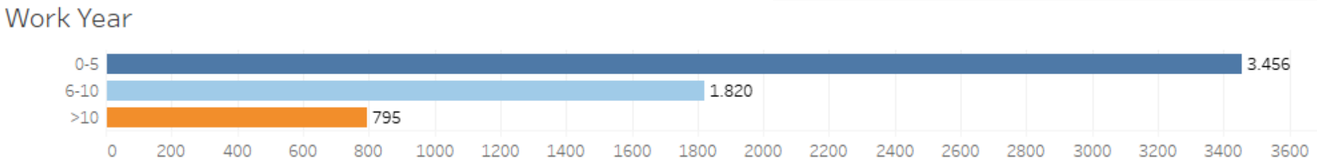
- Another feature that is not the key feature but still have an impact on the prediction models are **income type, flagging own a realty or car, and education level**.



Majority of Clients are Middle Age Adults, Middle Class Income, and in the Early Stage of Their Career





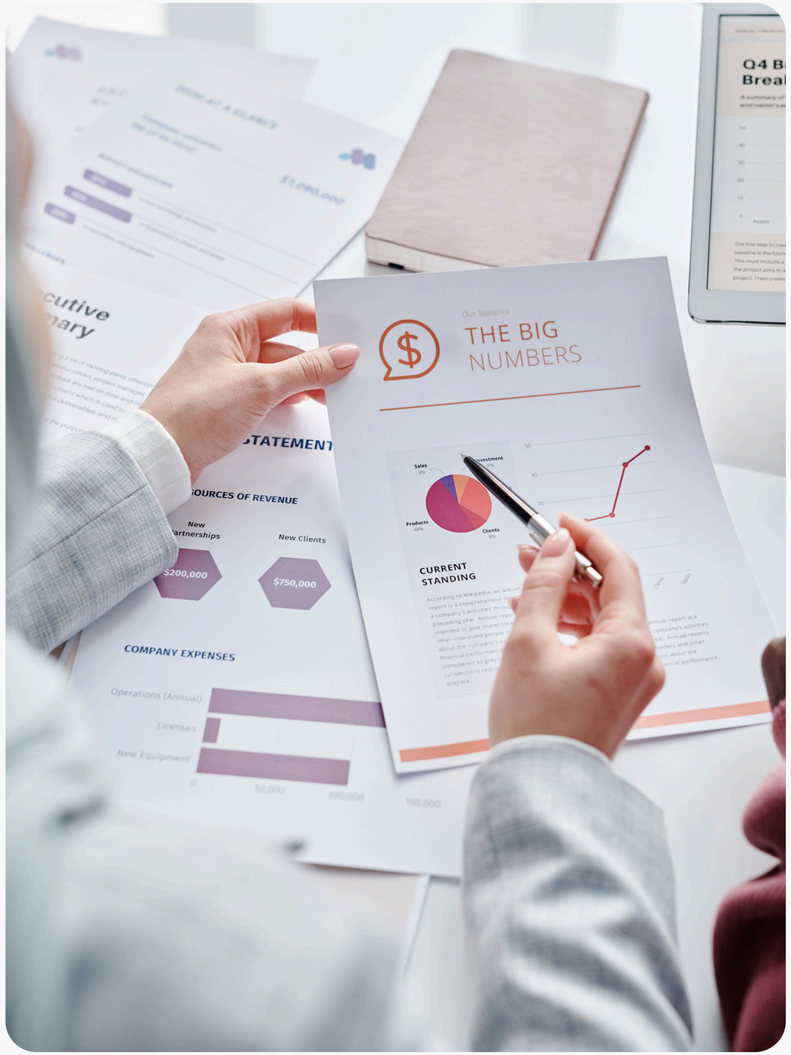


- The largest age group of clients that uses credit cards are 29-40 years and followed by the age group of 41-58 which indicate that clients that are using credit cards are mostly in the age group of 29-58 which tends to be financially established and mature in financial knowledge.

- Majority of the income amount of clients that have used credit cards are in the range of 100k-200k and 200k-300k which indicate most of the clients are middle class income.

- Majority of the clients working experiences are in the 0-5 years and 5-10 years. Which indicates most clients are in their early stage of their career.

Recommendations



Focus on the Key Feature:

- In accepting clients applications focus on key features previously mentioned
- Higher income, Older age, long work experience, all these indicate their capability to pay, financial maturity, and income stability all of which lower the risk of defaulting

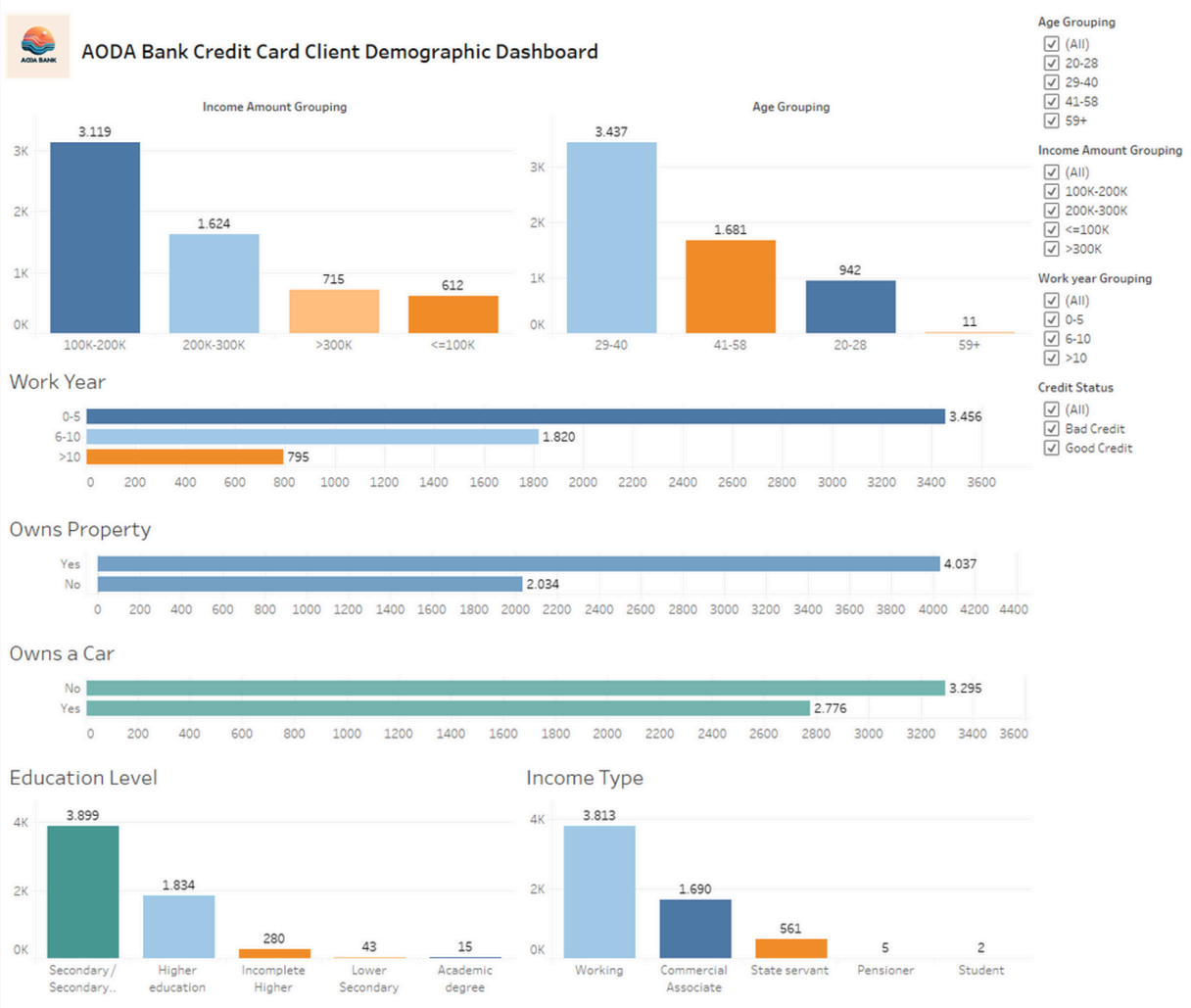
Check on Supporting Feature:

- Additional checking might be necessary to the supporting feature (Income type, flagging own a realty or car, and education level)
- These feature helps in deciding a person credit worthiness such as if a client owns a realty it indicates that they have the capacity to make a credit and also indicate there is a collateral if they fail to pay their credit.

Cater to Clients Needs:

- Based on the previous EDA the majority of the clients are still in the early career and in their middle age adults, and middle class income. Therefor the bank should cater to a middle tier credit card or credit builder credit cart for those with little to none credit history

Dashboard



Dashboard Link

Thank you

Appendices

Work Sheets and Dashboard

Data Sets

Account Records Link

Credit Records Link

Data Processing

Google Colab Link

Dashboard

Tableau Link